

## Concept Review

# Potentiometers

### Why Use Potentiometers?

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Potentiometers measure rotational or linear position. Potentiometers are used in many consumer products as an intuitive, tactile interface for users to adjust continuous values. We usually come across them in the form of dials or sliders, such as those used for adjusting volume or brightness.

## Rotary Potentiometers

A rotary potentiometer, or pot, is a manually controlled variable resistor. See the example shown in Figure 1. It typically consists of an exposed shaft, three terminals (A, W, and B), an encased internal resistive element shaped in a circular pattern, and a sliding contact known as a wiper. By rotating the shaft, the internal wiper makes contact with the resistive element at different positions, causing a change in resistance when measured between the center terminal (W) and either of the side terminals (A or B). The total resistance of the potentiometer can be measured by clamping a multimeter to terminals A and B.

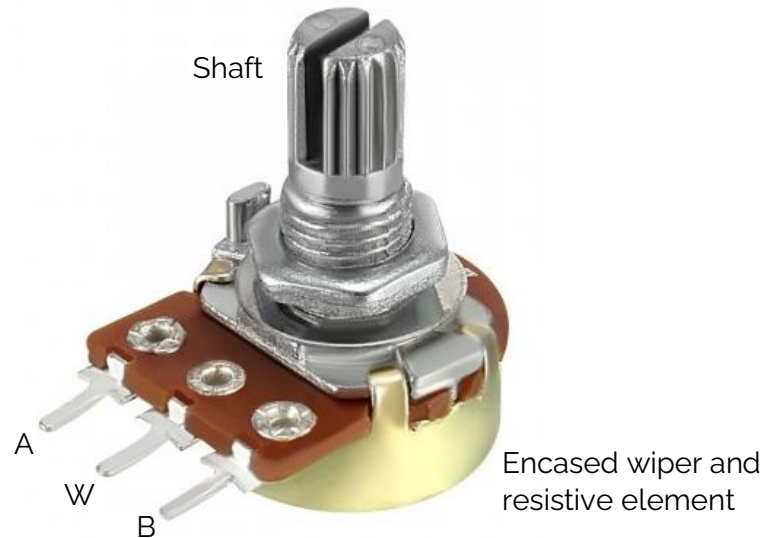


Figure 1: Schematic diagram of a potentiometer

A schematic diagram of the voltage dividing characteristic of a potentiometer is illustrated in Figure 2. By applying a known voltage  $V_{AB}$  between terminals A and B, voltage is divided between terminals AW and WB where,

$$V_{AB} = V_{AW} + V_{WB}$$

When connected to an external shaft, a rotary potentiometer can measure absolute angular displacement. By applying a known voltage to the outside terminals of the pot, we can determine the position of the sensor based on the output voltage  $V_{AW}$  or  $V_{WB}$  which will be directly proportional to the position of the shaft. One of the advantages of using a potentiometer as an absolute sensor is that after power loss, position information is retained since the resistance of the pot remains unchanged. While pots are an effective way to obtain a unique position measurement, caution must be used since their signal output may be discontinuous. That is, after a few revolutions potentiometers may reset their signal back to zero. Another disadvantage of most pots is that they have physical stops that prevent continuous shaft rotation. Drag forces between the resistive element the encased wiper must also be overcome by the mechanical system they are attached to, and wear and tear is an important consideration based on the life-cycle of the system they are used in.

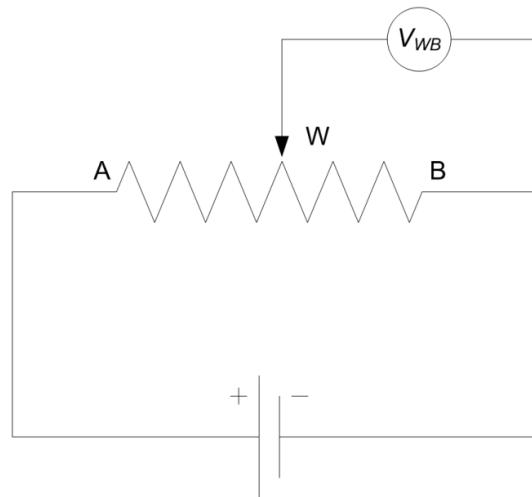


Figure 2: Schematic diagram of the voltage dividing characteristic of a rotary potentiometer.

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